

SMTP Challenge – Incident Analysis Report

Scenario:

During the analysis of a suspicious SMTP traffic capture (smtpchallenge.pcap), we investigated the communications of an insider, Ann, who was leaking sensitive information to her secret boyfriend. The following steps and findings detail how the investigation was conducted.

Tools Used

Wireshark (SMTP filtering, Follow → TCP Stream)

NetworkMiner (file extraction)

Base64 decoder (online or base64 -d)

md5sum / md5 (hash calculation)

Q1 — What is the email address of Ann's secret boyfriend?

21	51.386591	192.168.1.18	192.168.1.38	Syslog	278	KEPN.WARNING: Oct 18 12:13:46 gateway kernel: [46789.154281] FIREWALL:BLOCKEDDN=eth3 OUT= MAC=ffff:ffff:ffff:00:21:70:4d:4f:ae:00:00
22	54.059334	192.168.1.159	192.168.1.255	NBNS	92	Name query NB WORKGROUP=
23	54.060554	192.168.1.18	192.168.1.38	Syslog	278	KEPN.WARNING: Oct 18 12:13:47 gateway kernel: [46789.909817] FIREWALL:BLOCKEDDN=eth3 OUT= MAC=ffff:ffff:ffff:00:21:70:4d:4f:ae:00:00
24	54.080286	192.168.1.159	192.168.1.255	NBNS	92	Name query NB WORKGROUP=
25	54.080745	192.168.1.18	192.168.1.38	Syslog	278	KEPN.WARNING: Oct 18 12:13:48 gateway kernel: [46790.655862] FIREWALL:BLOCKEDDN=eth3 OUT= MAC=ffff:ffff:ffff:00:21:70:4d:4f:ae:00:00
26	57.555687	192.168.1.159	192.168.1.255	BROWSER	216	Get Backup List Request
27	57.556116	192.168.1.159	192.168.1.255	NBNS	92	Name query NB WORKGROUP=
28	57.556886	192.168.1.18	192.168.1.38	Syslog	280	KEPN.WARNING: Oct 18 12:13:50 gateway kernel: [46793.483893] FIREWALL:BLOCKEDDN=eth3 OUT= MAC=ffff:ffff:ffff:00:21:70:4d:4f:ae:00:00
29	57.557895	192.168.1.18	192.168.1.38	Syslog	278	KEPN.WARNING: Oct 18 12:13:50 gateway kernel: [46793.484878] FIREWALL:BLOCKEDDN=eth3 OUT= MAC=ffff:ffff:ffff:00:21:70:4d:4f:ae:00:00
30	58.303931	192.168.1.159	192.168.1.255	NBNS	92	Name query NB WORKGROUP=
31	58.304870	192.168.1.18	192.168.1.38	Syslog	278	KEPN.WARNING: Oct 18 12:13:51 gateway kernel: [46794.151573] FIREWALL:BLOCKEDDN=eth3 OUT= MAC=ffff:ffff:ffff:00:21:70:4d:4f:ae:00:00
32	59.058338	192.168.1.159	192.168.1.255	NBNS	92	Name query NB WORKGROUP=
33	59.059661	192.168.1.18	192.168.1.38	Syslog	278	KEPN.WARNING: Oct 18 12:13:52 gateway kernel: [46794.906877] FIREWALL:BLOCKEDDN=eth3 OUT= MAC=ffff:ffff:ffff:00:21:70:4d:4f:ae:00:00
34	61.080942	192.168.1.159	192.168.1.255	NBNS	92	Name query NB WORKGROUP=
35	61.081889	192.168.1.18	192.168.1.38	Syslog	278	KEPN.WARNING: Oct 18 12:13:55 gateway kernel: [46797.655856] FIREWALL:BLOCKEDDN=eth3 OUT= MAC=ffff:ffff:ffff:00:21:70:4d:4f:ae:00:00
36	62.554398	192.168.1.159	192.168.1.255	NBNS	92	Name query NB WORKGROUP=
37	62.555642	192.168.1.18	192.168.1.38	Syslog	278	KEPN.WARNING: Oct 18 12:13:55 gateway kernel: [46798.488223] FIREWALL:BLOCKEDDN=eth3 OUT= MAC=ffff:ffff:ffff:00:21:70:4d:4f:ae:00:00
38	63.386568	192.168.1.159	192.168.1.255	NBNS	92	Name query NB WORKGROUP=
39	63.387754	192.168.1.18	192.168.1.38	Syslog	278	KEPN.WARNING: Oct 18 12:13:56 gateway kernel: [46799.151924] FIREWALL:BLOCKEDDN=eth3 OUT= MAC=ffff:ffff:ffff:00:21:70:4d:4f:ae:00:00
40	63.395245	192.168.1.18	192.168.1.38	Syslog	230	KEPN.WARNING: Oct 18 12:13:59 gateway kernel: [46801.837528] FIREWALL:BLOCKEDDN=eth3 OUT= MAC= SHA-18.1.1.18 DST=18.1.1.255 LEN=76 TOS=
41	67.554211	VMware_08:00:0e:14	VMware_08:00:0e:14	ARP	42	Who has 192.168.1.18? Tell 192.168.1.18
42	67.554478	VMware_08:00:0e:14	VMware_08:00:0e:14	ARP	42	192.168.1.18 is at 08:00:c1:29:c8:09:0a
43	77.411223	192.168.1.18	192.168.1.18	NTP	80	NTP Version 4, client
44	77.411639	192.168.1.18	192.168.1.18	NTP	98	NTP Version 4, server
45	80.183237	192.168.1.159	192.168.1.255	BROWSER	255	Host Announcement ANN-LAPTOP, Workstation, Server, NT Workstation
46	80.184655	192.168.1.18	192.168.1.38	Syslog	280	KEPN.WARNING: Oct 18 12:15:13 gateway kernel: [46815.939471] FIREWALL:BLOCKEDDN=eth3 OUT= MAC=ffff:ffff:ffff:00:21:70:4d:4f:ae:00:00
47	82.312611	Dell_4d:4f:ae	Broadcast	ARP	42	Who has 192.168.1.18? Tell 192.168.1.159
48	82.312952	VMware_08:00:0e:14	Dell_4d:4f:ae	ARP	42	192.168.1.18 is at 08:00:c1:29:c8:09:0a
49	82.313441	192.168.1.159	18.1.1.28	DNS	72	Standard query Recv=9 A smtp.aol.com
50	82.407894	VMware_08:00:0e:14	VMware_08:00:0e:14	ARP	42	Who has 192.168.1.18? Tell 192.168.1.18
51	82.407200	VMware_08:00:0e:14	VMware_08:00:0e:14	ARP	42	192.168.1.18 is at 08:00:c1:29:c8:09:0a
52	82.678216	18.1.1.28	192.168.1.159	DNS	299	Standard query response Recv=9 A smtp.aol.com CNAME smtp.cs.com A 64.12.182.142 A 64.12.194.119 A 205.188.159.148 A 205.188.249.185 A 6
53	82.707978	192.168.1.159	64.12.182.142	TCP	82	1836 -> 587 [EST] Seq=8 Win=64248 Len=0 MSS=1460 'ACK_RECV'
54	82.817457	64.12.182.142	192.168.1.159	TCP	58	587 -> 1836 [SYN, ACK] Seq=8 Ack=1 Win=64248 Len=0 MSS=1460
55	82.822388	192.168.1.159	64.12.182.142	TCP	54	1836 -> 587 [ACK] Seq=1 Ack=1 Win=64248 Len=0
56	82.988897	64.12.182.142	192.168.1.159	SMTP	134	S: 228 cia-mc08-mx.aol.com ESMTP mail_cia-mc08.1; Sat, 18 Oct 2009 15:35:16 -0400
57	82.998439	192.168.1.159	64.12.182.142	SMTP	78	C: EHLO annlaptop

Method:

Filter packets by smtp in Wireshark and inspect the RCPT TO entry or follow the relevant SMTP TCP stream.

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220 cia-mc07.mx.aol.com ESMTP mail_cia-mc07.1; Sat, 10 Oct 2009 15:37:56 -0400
EHLO annlaptop
250-cia-mc07.mx.aol.com host-69-140-19-190.static.comcast.net
250-AUTH=LOGIN PLAIN XAOL-UAS-MB
250-AUTH LOGIN PLAIN XAOL-UAS-MB
250-STARTTLS
250-CHUNKING
250-BINARYMIME
250-X-AOL-FWD-BY-REF
250-X-AOL-DIV_TAG
250-X-AOL-OUTBOX-COPY
250 HELP
AUTH LOGIN
334 VxNlcm5hbWU6
c25lYm5kZmZlY291
334 UGFzc3dvcmQ6
VTU4cjAwbHo=
235 AUTHENTICATION SUCCESSFUL
MAIL FROM: <sneakyg33k@aol.com>
250 OK
RCPT TO: <mistersecretx@aol.com>
250 OK
DATA
354 START MAIL INPUT, END WITH "." ON A LINE BY ITSELF
Message-ID: <001101ca49ae93e45b059f01a8c0@annlaptop>
From: "Ann Dercover" <sneakyg33k@aol.com>
To: <mistersecretx@aol.com>
Subject: rendezvous
Date: Sat, 10 Oct 2009 07:38:10 -0600
MIME-Version: 1.0
Content-Type: multipart/mixed;
boundary="====_NextPart_000_000D_01CA497C.9DEC1E70"
X-Priority: 3
X-MSMail-Priority: Normal
X-Mailer: Microsoft Outlook Express 6.00.2900.2180
X-MimeOLE: Produced By Microsoft MimeOLE V6.00.2900.2180

This is a multi-part message in MIME format.

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Answer: mistersecretx@aol.com

Q2 — What is Ann's email password?

Method:

No.	Time	Source	Destination	Protocol	Length	Info
53	82.707578	192.168.1.159	64.12.102.142	TCP	62	1836 → 587 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM
54	82.817457	64.12.102.142	192.168.1.159	TCP	58	587 → 1836 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460
55	82.822288	192.168.1.159	64.12.102.142	TCP	54	1836 → 587 [ACK] Seq=1 Ack=1 Win=64240 Len=0
56	82.908997	64.12.102.142	192.168.1.159	SMTP	134	S: 220 cia-mc06.mx.aol.com ESMTP mail_cia-mc06.1; Sat, 10 Oct 2009 15:35:16 -0400
57	82.998439	192.168.1.159	64.12.102.142	SMTP	70	C: EHLO annlaptop
58	82.998926	64.12.102.142	192.168.1.159	TCP	54	587 → 1836 [ACK] Seq=81 Ack=17 Win=64240 Len=0
59	83.107523	64.12.102.142	192.168.1.159	SMTP	385	S: 250-cia-mc06.mx.aol.com host-69-140-19-190.static.comcast.net AUTH=LOGIN PLAIN XAOL-UAS-MB AUTH LOGIN
60	83.109070	192.168.1.159	64.12.102.142	SMTP	66	C: AUTH LOGIN
61	83.110259	64.12.102.142	192.168.1.159	TCP	54	587 → 1836 [ACK] Seq=332 Ack=29 Win=64240 Len=0
62	83.220242	64.12.102.142	192.168.1.159	SMTP	72	S: 334 VxNlcm5hbWU6
63	83.221800	192.168.1.159	64.12.102.142	SMTP	80	C: User: c25lYm5kZmZlY291
64	83.221908	64.12.102.142	192.168.1.159	TCP	54	587 → 1836 [ACK] Seq=350 Ack=55 Win=64240 Len=0
65	83.331342	64.12.102.142	192.168.1.159	SMTP	72	S: 334 UGFzc3dvcmQ6
66	83.331953	192.168.1.159	64.12.102.142	SMTP	68	C: Pass: VTU4cjAwbHo=
67	83.332382	64.12.102.142	192.168.1.159	TCP	54	587 → 1836 [ACK] Seq=368 Ack=69 Win=64240 Len=0
68	83.462637	64.12.102.142	192.168.1.159	SMTP	85	S: 235 AUTHENTICATION SUCCESSFUL
69	83.465436	192.168.1.159	64.12.102.142	SMTP	87	C: MAIL FROM: <sneakyg33k@aol.com>
70	83.466809	64.12.102.142	192.168.1.159	TCP	54	587 → 1836 [ACK] Seq=399 Ack=102 Win=64240 Len=0
71	83.578844	64.12.102.142	192.168.1.159	SMTP	62	S: 250 OK
72	83.579690	192.168.1.159	64.12.102.142	SMTP	83	C: RCPT TO: <sec558@gmail.com>
73	83.580124	64.12.102.142	192.168.1.159	TCP	54	587 → 1836 [ACK] Seq=407 Ack=131 Win=64240 Len=0
74	83.697311	64.12.102.142	192.168.1.159	SMTP	62	S: 250 OK
75	83.698197	192.168.1.159	64.12.102.142	SMTP	68	C: DATA
76	83.698687	64.12.102.142	192.168.1.159	TCP	54	587 → 1836 [ACK] Seq=415 Ack=137 Win=64240 Len=0
77	83.808024	64.12.102.142	192.168.1.159	SMTP	110	S: 354 START MAIL INPUT, END WITH "." ON A LINE BY ITSELF
78	83.818602	192.168.1.159	64.12.102.142	SMTP	1482	C: DATA fragment, 1348 bytes
79	83.812130	64.12.102.142	192.168.1.159	TCP	54	587 → 1836 [ACK] Seq=471 Ack=1485 Win=64240 Len=0
80	83.812602	192.168.1.159	64.12.102.142	SMTP/IMF	59	From: "Ann Dercover" <sneakyg33k@aol.com>, subject: lunch next week, (text/plain) (text/html) .
81	83.812258	64.12.102.142	192.168.1.159	TCP	54	587 → 1836 [ACK] Seq=471 Ack=1490 Win=64240 Len=0
82	84.148051	64.12.102.142	192.168.1.159	SMTP	62	S: 250 OK
83	84.149429	192.168.1.159	64.12.102.142	SMTP	68	C: QUIT
84	84.149822	64.12.102.142	192.168.1.159	TCP	54	587 → 1836 [ACK] Seq=479 Ack=1496 Win=64240 Len=0
85	84.259720	64.12.102.142	192.168.1.159	SMTP	83	S: 221 SERVICE CLOSING CHANNEL
86	84.264151	192.168.1.159	64.12.102.142	TCP	54	1836 → 587 [ACK] Seq=1496 Ack=589 Win=63733 Len=0
87	84.278073	192.168.1.159	64.12.102.142	TCP	54	1836 → 587 [FIN, ACK] Seq=1496 Ack=589 Win=63733 Len=0
88	84.278014	64.12.102.142	192.168.1.159	TCP	54	587 → 1836 [ACK] Seq=509 Ack=1497 Win=64239 Len=0

Locate the SMTP authentication packet (packet 66 in this capture). The authentication string is Base64 encoded — copy the Base64 token and decode it

Decode from Base64 format
Simply enter your data then push the decode button.

NTU4cjAwbHo=

For encoded binaries (like images, documents, etc.) use the file upload form a little further down on this page.

UTF-8 Source character set.

☐ Decode each line separately (useful for when you have multiple entries).

☒ Live mode OFF Decodes in real-time as you type or paste (supports only the UTF-8 character set).

< DECODE > Decodes your data into the area below.

558r00lz

Answer: (decoded from packet 66; Base64 → Answer: 558r00lz

Q3 — What is the name of the file that Ann sent to her secret lover?

Method:

Inspect the SMTP DATA portion in the relevant stream (packet 140) and look for the Content-Disposition / filename field.

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i sweetheart! Bring your fake passport and a bathing suit. Address =
ttached. love, Ann
-----_NextPart_001_000E_01CA497C.9DEC1E70
Content-Type: text/html;
    charset="iso-8859-1"
Content-Transfer-Encoding: quoted-printable

!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.0 Transitional//EN">
HTML<<HEAD>
META http-equiv=3DContent-Type content=3D"text/html; =
    charset=3Diso-8859-1">
META content=3D"MSHTML 6.00.2900.2853" name=3DGENERATOR>
STYLE<</STYLE>
/HEAD>
BODY bgcolor=3D#ffffff>
DIV><FONT face=3DArial size=3D2>Hi sweetheart! Bring your fake passport =
    nd a=20
athing suit. Address attached. love, Ann</FONT></DIV></BODY></HTML>

-----_NextPart_001_000E_01CA497C.9DEC1E70--
-----_NextPart_000_000D_01CA497C.9DEC1E70
Content-Type: application/octet-stream;
    name="secretrendezvous.docx"
Content-Transfer-Encoding: base64
Content-Disposition: attachment;
    filename="secretrendezvous.docx"

```

Answer: secretrendezvous.docx

Q4 — In what country will Ann meet with her secret lover?

Method:

Extract the .docx attachment from the SMTP stream (save the TCP stream as .eml, then open or extract the attachment with NetworkMiner or an email client). Open the Word document to read the embedded content — the highlighted map shows the meeting location.

Answer: Mexico

Q5 — What is the MD5 value of the attachment Ann sent?

Method:

After extracting secretrendezvous.docx, compute its MD5 hash with md5sum secretrendezvous.docx (or view File Details in NetworkMiner).

Answer: 9e423e11db88f01bbff81172839e1923

Summary / Findings

The SMTP capture revealed cleartext-leaked data, including the recipient address (mistersecretx@aol.com), an attached document (secretrendezvous.docx), and network-exposed credentials (Base64 encoded in the SMTP AUTH exchange).

The attachment contains the meeting location (Mexico) and its MD5 hash is 9e423e11db88f01bbff81172839e1923.

Action items: reset the exposed credentials immediately, notify affected parties, and perform a deeper review of outbound email monitoring and DLP controls to prevent further data exfiltration.